

Online Quiz System - Project Requirements and Design

1. Functional Requirements:

- Users can register and log in.
- Users can start a new quiz.
- The app generates questions dynamically using the REST Countries API.
- Users can select quiz categories (e.g., capitals, flags).
- Each quiz contains a fixed number of random questions.
- The app provides feedback on answers.
- The application stores and displays user scores and statistics.
- Admins can view analytics and manage users.

2. Non-Functional Requirements:

- The app should respond within 2 seconds for any button press.
- The response from the app server must be cached to reduce API calls.
- The application must support at least 100 concurrent users.
- The backend should be built using Spring Boot.
- The frontend should be built with Thymeleaf.
- The app must be secure, storing passwords in hashed form.
- The system should log errors and user activities.

3. Use Cases

Use Case Name	Description	Actor
Register	New users sign up to the system.	User
Login	Registered users log into the system.	User
Start Quiz	User selects category and begins a quiz.	User
Answer Question	User selects an answer for a question.	User

Submit Quiz	User submits the quiz and sees results.	User
View Score History	Users can view past quiz results.	User
Manage Users	Admin can view and delete users.	Admin

4. Objects, Classes, and Relationships:

Main Classes and Attributes:

- User: id, username, email, password, role, registrationDate
- Quiz: id, userId, category, startTime, endTime, score
- Question: id, quizId, questionText, correctAnswer, options[]
- CountryInfoService: fetches data from REST Countries API
- Answer: id, questionId, userAnswer, isCorrect
- Admin: will inherits from User

Relationships:

- One User can take many Quizzes → one-to-many
- One Quiz contains many Questions → one-to-many
- One Question has many Answers → one-to-many

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